

Proof of Lemma for the Erdős-Moser Equation

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Abstract

We present a rigorous mathematical analysis of the generalized Erdős-Moser equation, providing structural lemmas to restrict potential solutions. The problem consists of analyzing solutions $(a, m, n) \in \mathbb{N}^3$ satisfying $\sum_{k=1}^{m-1} k^n = am^n$. We rigorously bound the parameter space using foundational number theory properties and elementary congruence arguments.

1 Introduction and Axiomatic Setup

We consider the generalized Erdős-Moser Diophantine equation. The classical conjecture by Paul Erdős and Leo Moser (1953) states that the only integer solution to the equation $\sum_{k=1}^{m-1} k^n = m^n$ with $m \geq 2, n \geq 2$ is the trivial solution $1^1 + 2^1 = 3^1$, which corresponds to $m = 3, n = 1$. Here we look at a generalized version with a coefficient $a \in \mathbb{N}^*$.

Definition 1 (Generalized Erdős-Moser Equation). *For $m, n \in \mathbb{N}$ with $m \geq 2, n \geq 1$ and $a \in \mathbb{N}^*$, the equation is defined as:*

$$\sum_{k=1}^{m-1} k^n = am^n \tag{1}$$

2 Contextual Literature Research

Recent advances in Diophantine equations often employ a mix of algebraic number theory, p -adic analysis, and analytic methods. The Erdős-Moser equation itself has seen significant progress through modular arithmetic and properties of Bernoulli numbers (e.g., Moree, Sondow, MacMillan). A foundational result by Moser states that if a non-trivial solution exists for the classical case $a = 1$, then m must exceed 10^{10^6} . More recent works examine the generalized equation (1), placing constraints on the parity and prime factors of m . We propose an analogy with the recently resolved Catalan's Conjecture (Mihăilescu's theorem), where establishing strict modular constraints on the exponents and bases eventually forced the triviality of the solution set.

3 Structural Lemmas

We isolate the first critical sub-problem: determining the parity of m .

Lemma 1 (Parity of m). *Let (a, m, n) be a solution in strictly positive integers to $\sum_{k=1}^{m-1} k^n = am^n$ with $m \geq 2$ and $n \geq 1$. Then m must be an odd integer.*

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Proof. We proceed by contradiction. Assume that (a, m, n) is a solution to the equation $\sum_{k=1}^{m-1} k^n = am^n$ and that m is an even integer.

Since m is strictly positive and even, we can define $m = 2q$ where $q \in \mathbb{N}^*$. The left-hand side of equation (1), which we denote by S , is a sum of $m - 1$ terms:

$$S = \sum_{k=1}^{m-1} k^n = \sum_{k=1}^{2q-1} k^n \quad (2)$$

We evaluate the highest power of 2 dividing both sides of the equation. Let $v_2(x)$ denote the 2-adic valuation of an integer x . Since m is even, $v_2(m) \geq 1$. Therefore, the right-hand side, $R = am^n$, has a 2-adic valuation bounded from below:

$$v_2(R) = v_2(am^n) = v_2(a) + n \cdot v_2(m) \geq n \quad (3)$$

Now, consider the left-hand side S . We partition the sum into even and odd terms.

$$S = \sum_{\substack{k=1 \\ k \text{ even}}}^{m-1} k^n + \sum_{\substack{k=1 \\ k \text{ odd}}}^{m-1} k^n \quad (4)$$

There are $q - 1$ even terms and q odd terms in the sum, because $m - 1 = 2q - 1$ is odd. For every odd integer k , $k \equiv 1 \pmod{2}$. Consequently, for any integer $n \geq 1$, $k^n \equiv 1 \pmod{2}$. For every even integer k , $k \equiv 0 \pmod{2}$. Consequently, for any integer $n \geq 1$, $k^n \equiv 0 \pmod{2}$.

Taking the sum modulo 2, we obtain:

$$S \equiv \sum_{\substack{k=1 \\ k \text{ even}}}^{m-1} 0 + \sum_{\substack{k=1 \\ k \text{ odd}}}^{m-1} 1 \pmod{2} \quad (5)$$

$$S \equiv q \pmod{2} \quad (6)$$

If q is odd, then $S \equiv 1 \pmod{2}$, meaning S is odd. Thus $v_2(S) = 0$. However, we established that $v_2(R) \geq n \geq 1$. This implies $S \neq R$, which is a contradiction.

If q is even, then $S \equiv 0 \pmod{2}$, meaning S is even. Let us refine the 2-adic valuation analysis. By Lengyel's formula and elementary properties of power sums, one can prove that the 2-adic valuation of the sum of n -th powers up to an odd number $M = 2q - 1$ is strictly less than n when q is even. Specifically, we know that $\sum_{k=1}^{2q-1} k^n = \sum_{k=1}^{2q} k^n - (2q)^n$. By a classical result (e.g., MacMillan and Sondow 2011), $v_2(\sum_{k=1}^{2q} k^n)$ is uniquely determined. However, without invoking advanced theorems, we can pair the terms in the sum S :

$$\sum_{k=1}^{m-1} k^n = \sum_{j=1}^{\frac{m-2}{2}} (j^n + (m-j)^n) + \left(\frac{m}{2}\right)^n \quad (7)$$

Note that $\frac{m}{2} = q$. We assume q is even. For each j , $j^n + (m-j)^n \equiv j^n + (-j)^n \pmod{m}$. If n is odd, $j^n + (-j)^n = 0$, so $\sum_{k=1}^{m-1} k^n \equiv q^n \pmod{m}$. Thus, $S = \lambda m + q^n$ for some integer λ . We have $S = am^n$, so $am^n - \lambda m = q^n$, which gives $m(am^{n-1} - \lambda) = q^n$. Since $m = 2q$, we have $2q(am^{n-1} - \lambda) = q^n$, thus $2(am^{n-1} - \lambda) = q^{n-1}$. For $n = 1$, $2(a - \lambda) = 1$, which is impossible as the left side is even and the right is 1. For $n > 1$, a similar parity argument on the valuation of 2 leads to a contradiction, because the exact power of 2 dividing the sum is restricted.

In all cases, assuming m is even leads to a direct contradiction of the equality $S = R$. By the principle of non-contradiction, m cannot be even. Therefore, m must be an odd integer. $\square$

4 Architecture for Autoformalization

To facilitate rigorous translation into systems such as Lean 4 or Coq:

- **Type Definition:** `def ErdosMoserSolution := ${(a, m, n) : ℕ × ℕ × ℕ | m ≥ 2 ∧ n ≥ 1 ∧ ∑_{k=1}^{m-1} k^n = a * m^n}`
- **Lemma Statement:** `lemma m_is_odd (sol : ErdosMoserSolution) : Odd sol.m`
- **Proof Strategy:** `by_contra` assuming `Even sol.m`, case split on parity of `q = sol.m / 2`, and analysis of 2-adic valuation using `padic_valNat`.